MID-TERM EXAMINATION
(Course Name B.Tech ECE) (Semester 3rd)
(October, 2023) OFF LINE mode

Subject Code: BEC 203	**Subject: Signals & Systems**
Time : 1 ½Hours	**Maximum Marks : 30**
Note: Q. 1 is compulsory.	

Q1		**(2.5*4)**	
	(a) Explain Deterministic signals and Random Signals .		
	(b) Difference Between Continuous -time unit step function and Discrete-time unit step function.		
	(c) What are the properties of Linear Time Invariant Systems.(Explain any 2 with examples)		
	(d) Check wether the systems are Periodic or Aperiodic-: I. $X(t)=\sin 20\pi t$ II. $X(t)=\cos 8t+\sin\sqrt{3}$		
Q2	**(Attempt any Two Parts) UNIT-1**	**(5,5)**	
	(a) Explain the terms-: I. Principle of Homogeneity II. Superposition Principle		
	(b) Check wether the systems are-: I. $X(t)=Sin20\pi t$ — Periodic/Aperiodic II. $Y(t)=\text{Log}[x(t)]$ — Static/Dynamic III. $Y(t)=e^{x(t)}$ — Causal/Non-Causal IV. $Y(t)=x(3-t)+x(t-8)$ — Causal/Non-Causal V. $Y(t)=x(t-6)$ — Static/Dynamic		
	(c) Draw the waveforms of the following-: I. $X_1(t)=u(t+2)$ II. $X_3(t)=u(-t)$		
Q3	**(Attempt any Two Parts) UNIT-2**	**(5,5)**	
	(a) Mention any 3 properties of Periodic Signals .		
	(b) Find the Fourier Transform of-: $X(t)=\cos(w_c t)\,u(t)$		
	(c) Explain the below mentioned properties with proof-: I. Linearity II. Time Shifting III. Frequency Shifting IV. Time Reversal		